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Resident space object (RSO) attitude and optical property estimation from space-based light curves

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Abstract

With the increase in the number of objects orbiting Earth, Space Situational Awareness (SSA) has becoming an important area of research in the space sector. Currently most sensors that contribute to SSA are large dedicated optical or radar stations, such as space fence (Pechkis, et al., 2015). With the increase in low resolution sensors in LEO there is a growing potential to utilize these to augment current SSA efforts. Star trackers are readily available and used in space for attitude determination, with recent work performed to demonstrate the benefit of using spaced-based optical measurements for Resident Space Object (RSO) detection. In this paper, we describe the interpretation of space-based measurement for light curve of an RSOs to estimate the RSOs shape, attitude and optical properties. In this model, two Bidirectional Reflectance Distribution Functions (BDRF's) are compared, namely a defined facet model and an anthropic Phong model. From the initial results an RSOs shape, attitude, optical properties can be estimated with basic a-priory information on the shape of the RSO with both models.

Keywords: Space Situational Awareness; Light Curve Inversion; Attitude Estimation; Bidirectional Reflectance Distribution; Resident Space Object

1. Introduction

Damage to space systems has a significant and immediate impact on technologies we rely on every day, including navigation, communication, resource management and weather forecasting. The 2009 collision of Iridium 33 and Kosmos 2251 destroyed both satellites and created thousands of debris, which resulted in a loss of service (Baur, et al., 2014). As such, it is imperative to enhance technologies and further develop capabilities that identify both active satellites and inactive resident space objects (RSO's) such as debris. According to NASA (Clemens et al., 2019), there are more than 500,000 objects above 1 cm size in Earth's orbit. It is still unclear how many hundreds of thousands of uncatalogued objects pose threats to space assets, such as the International Space Station. While there are continuing and collaborative.

efforts among various space agencies and research communities to monitor the resident space objects from the ground and on-orbit, there still remains many uncertainties in RSO numbers and characteristics.

In (Lambour, et al., 2000), we demonstrated RSO identification from space-based observations using low-resolution imager spacecraft such as the Fast Auroral Imager (FAI) onboard the CASSIOPE satellite. A sample FAI image compared to a simulated image is illustrated in Fig. 1, with the RCM satellites being labeled in both images. In order to interpret the low-resolution spaceborne images further study of the light curve is required to infer the parameters of the RSO. A light curve is a representation of the brightness of an object as a function of time. In the context of this paper the parameters include; optical properties, shape and attitude of the RSO. Knowledge of

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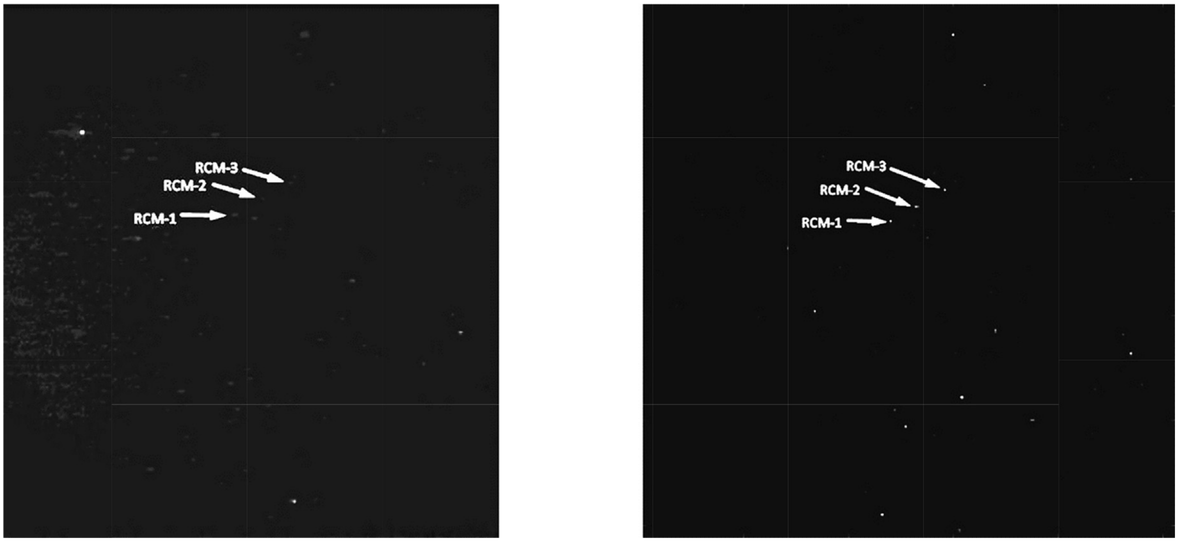

Fig. 1. Sample Scaled Images of RCM constellation from FAI on-orbit observation (left), image credit (University of Calgary, Department of Physics and Astronomy, 2019), simulated images of the same RCM constellation using SBOIS (Right) take on June15, 2019, 23:36:05.

these parameters gives satellite operators and different agencies more information on targeted debris and active objects giving more information than just position information increasing SSA knowledge. Knowledge of these parameters versus just position information gives the required information for performing active debris removal, improved RSO correlation, and confirming that satellite ahead to current international regulations. Recent research has been focused on techniques that allow the interpretation of light curves from the images of debris objects, such as in (Gasdia, 2016) (Gasdia, et al., 2017) (Bernander, 2014) (Payne, et al., 2009). Other research in this field focuses on complex modeling techniques to simulate space-based surveillance images and photometric light curves (Fulcoy, et al., 2012) (Kaasalainen, 2001). These algorithms provide a baseline for the current study for image simulation of the space-based images for RSO identification, tracking, and characterization. The majority of other image processing research focuses on ground-based observations with limited applicability for space-based images. Space-based research is still in its early stage; the design and implementation processes require further optimization. In this paper, we address the rising need for better SSA by comparing two different bidirectional reflectance distribution functions (BRDF) in a simulated environment. The goal of this research is to the development a suitable algorithm to estimate parameters from light curves extracted from low-resolution space-based images. This is accomplished through the generation of simulated light curves with known parameters then comparing the generated light curve, commonly called synthetic light curve, to the observed light curve of the object.

Light curves are used to interpret various physical parameters of the object under investigation, mentioned previously. In the case of RSO characterization, the frequency analysis of light curves is often used to estimate the rate of rotation (also referred to as spin-rate) of the object. In (Lambour, et al., 2000) (and many articles with similar analysis), the authors present an overview of how light curve analysis is used for RSO characterisation using low- resolution images. More generally, light curve inversion is a mathematical technique used to estimate different states of an objects from its brightness and brightness variation over a series of timed measurements, which provides more information than spin rate analysis.

Light curve inversion is frequently used for the detection of exoplanets and the shape and spin estimation of asteroids, such as in (Cowan & Agol, 2008) and (Clark, et al., 2020). To invert the light curve to determine more information than just the temporal spin rate of the object, simulated environments are commonly used. They determine how the variation of brightness corresponds to a change in parameters which are used to generate the simulated light curve. To perform a light curve inversion an optical image simulator, in the case of this paper the Space Based Optical Image Simulator (SBOIS), is used to replicate the brightness of the RSO with estimated states that consists of shape, attitude and optical properties. The complexity of the light curve inversion problem stems from the brightness giving non-unique solutions for state estimation, where hundreds to thousands of solutions are plausible for one brightness measurement. To overcome the challenge of non unique answers a time series of brightness (the light curve) is used. The time series allows for higher derivatives of data to be used to reduce the number of potential solutions, or local optimum, present. The higher derivative data limits the solution space by ignoring solutions with large discrepancies in state parameters that would not be physically possible. One example of this is large attitude jumps that can sometime be larger than 30 degrees a second. It is unlikely to leave only the ideal solution, even after higher derivative fitting, requiring optimization to find better estimated states. It was demonstrated in previous research that Particle Swarm Optimization performs effectively for attitude estimation; thus, it is implemented in the current study to calculate the optimal state estimates (Snell, 2010). It is worthwhile to note that any optimization algorithm in theory can be implemented to retrieve an optimal state estimate. In this paper, the methodology for generation of simulated images, as well as, simulated environment used is described. Followed by the results and discussion of the simulation tests and concluded with a summary of the results and areas of future research mentioned.

2. Methodology

2.1. Simulated RSO images

In order to estimate RSO's physical properties; attitude, shape, and optical properties, from a time-series of images, light curve inversion is applied. It is implemented iteratively to check for optimal solutions by generating the simulated light curve, then updating the sim- ulated light curve until it matches the brightness of the observed light curve. Where the observed light curve represents the light curve extracted from real images of the object. The simulated light curve represents the light curve of the RSO generated in a simulated environment from estimated parameters. To generate the simulated light curves SBOIS was used. SBOIS was tailored to generate thousands of images, or in this case light curves, in an efficent manor which is then fed into an optimization algorithm for the state update. SBOIS is a simulated optical environment, fully coded in MATLAB. SBOIS is used to generate the Earth, Moon, Sun, Star Field, Zodiac Background Light and RSOs for space based instruments. Different types of environmental and instrument noise are included in the simulator in the form of shot noise, dark noise, Multi Pixel Radiation events, Point Spread Function and hot pixels.

The optical environment considers all light from RSO's to be attributed to reflected solar radiation, which is calculated with the relative brightness Equation (1) (McCue, et al., 1971).

$$m_v = \frac{148}{149} m_0 - 2.5 \log \frac{\rho A F(\varphi)}{R_2^{151}} \quad (1)$$

In equation (1); m_v and m_o refer to the visual magnitude of the target and reference source, respectively. ρ represents the reflectivity of the object. A represents the effective area of the target object. F (Φ) represents the phase function. R represents the distance between the observer and object.

Normally, only spherical RSO models are considered; therefore, only the phase angle is required. However, simulating objects as perfectly spherical reflectors does not give the different specular and diffuse reflections across the many surfaces of an RSO. Spherically modeling RSO's leads to the light curve being a smooth curve based on the distance to the object. This does not accurately represent real RSO light curves or allow for light curve inversion to be performed. To simulate accurate reflections and light curves off of a complex 3D RSO, facet models are used to replicate the RSO's reflective surfaces. A facet model of an RSO takes the 3D shape of the object and models it as a superposition of 2D surfaces called facets. The facets chosen have well known BRDF functions allowing for the complex 3D shape to be modeled from the sum of simpler 2D surfaces. With the implementations of facet models, the phase function becomes a function of more than just solar phase angle, Φ. This is shown in the rest of this paper by representing the phase function as F for both facet models examined. As SBOIS is not the focus of this paper more information on SBOIS is available upon request, with upcoming studies being performed on the comparison of SBOIS to STK's EOIR toolkit.

2.2. RSO facet models of bidirectional reflectance distribution function (BRDF)

To simulate the brightness of an RSO, its shape, size, and optical properties need to be accurately defined and modeled. As mentioned above for RSO orbit determination or detection algorithms, it is often modeled as a perfectly spherical reflecting object with a known radius. For light curve representation, more complex models of RSO's are required, where the 3D shape is modeled as a series of facets or 2D surfaces. Facet models allow for the estimation of the light reflected off of complex shapes by approximating the complex shape as many simpler surfaces with well known Bidirectional Reflectance Distribution Functions (BRDF) called facets. BRDFs calculate the amount of light that is reflected off of a facet in the direction of the observer. BRDFs calculates this from the observer, target, light source geometry, which in the case of observing RSOs represents: host, RSO, Sun respectively. BRDF also depends on the optical parameters of the facet such as reflectivity and diffuse vs specular reflection weighting. The different optical parameters that are used depend on the BRDF model that is used, with two examples of this being the Standard model and Anisotropic Phong model explored in this paper. Different BRDF models for the facets are available with the first facet models for satellites being introduced in the early 1970′s (Krag, 1974), (Hejduk, 2011). Table 1 illustrates the diffuse verses specular conditions for reflection and the respective BRDF model for each facet. Three facets are used in this paper to represent all 3D shapes; they represent the varying degrees of curvature across a facet (also outlined in Table 1.) Diffuse reflection refers to the reflection where the angle of incidence is not the angle of reflection. Specular, or mirror like, reflection happens when the angle of incidence is equal to the angle of reflection. Generally, specular reflections are orders of magnitude brighter than diffuse reflection. Both these reflections need to conserve the reflected radiation off of the surface; to represent this the Hejduk Weighting

Table 1
Standard Facet Dependent Phase Functions (Linares, et al., 2014).

Facet Type (Degrees of Curvature)	Sphere (2)	Flat Surface (0)	Cylinder (1)
Diffuse Phase Condition	None	ϕ_{obs} and ϕ_{sun} are the same sign	None
Diffuse Phase Equation F_{diff}	$\frac{2}{3\pi^2}[(\pi-\Phi)\cos(\Phi)+\sin(\Phi)]$	$\frac{1}{\pi}\sin(\phi_{obs})\sin(\phi_{sun})$	$\frac{\cos(\phi_{obs})\cos(\psi_{sun})}{4\pi}[(\pi-\theta)\cos(\theta)+\sin(\theta)]$
Specular Phase Condition	None	$\lvert\phi_{obs}-\phi_{sun}\rvert \le \frac{\Delta}{2} \& \lvert\theta_{sun}-\theta_{obs}\rvert = \pi$	$\lvert\phi_{obs}-\phi_{sun}\rvert \le \frac{\Delta}{2}$
Specular Phase Equation F_{spec}	$\frac{1}{4\pi^2}$	$\frac{4\cos(\Phi/2)}{\pi\Delta^2}$	$\frac{\cos(\Phi/2)\alpha(t)}{4\Delta}$

Coefficients, B_{spec} and B_{diff}, are used. These coefficients define the ratio of reflected light from specular and diffuse reflections, illustrated in equations (2) to (3) (Linares, et al., 2014). The viewing geometry for Table 1 is shown in Fig. 2.

In Table 1, Δ represents the reflected fan of light set to be the diameter of the sun 0.0093 rad at Earth's distance. α(t) is a time dependent function that calculates the integrated light across a chord of the sun's disk.

$$1 = BSpec + BDiff + Babs \tag{2}$$

$$F = FdiffBdiff + FSpecBSpec \tag{3}$$

Since the initial facet model for satellites were first developed in the 70′s, different BRDF models have been examined and implemented to represent the satellites in orbit. An exam- ple that has been widely accepted in recent years is the Anisotropic Phong model (Linares, et al., 2014) (Ashikhmin & Shirley, 2000). The Anisotropic Phong model uses empirically found values, n_u and n_v, which accounts for the reflection across the different axis of the facet. The Anisotropic Phong model is proposed for RSO's image simulation because it has the possibility of reducing the possible state estimations due to the reducing ambiguity in the facet orientation. The Phong model is a much more complex example, with a simplification for flat facets shown in equations (4) through (7). Flat facets sufficiently represent active satellites with large flat surfaces, such as solar panels and reflective antennas. For this current study, we have applied flat facet model to represent the RADARSAT Constellation Mission (RCM) satellites. Fig. 3 shows the observation geometry of the Phong model.

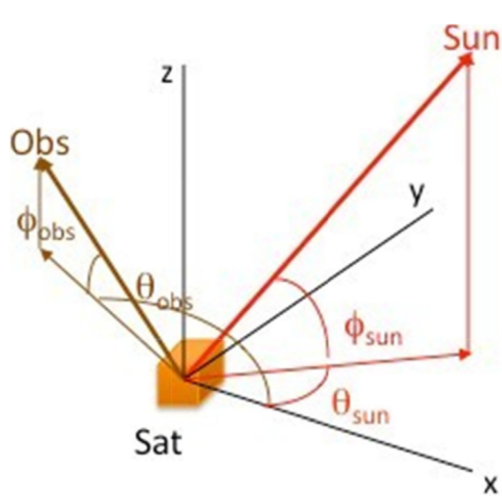

Fig. 2. Illumination Geometry (Pesce, et al., 2019).

$$F_{spec} = \frac{\sqrt{(n_u+1)(n_v+1)}}{8\pi} \times \frac{(u_n \cdot u_h)^{n_u(u_h \cdot u_u)^2 + n_v(u_h \cdot u_v)^2}}{u_n \cdot u_{sun} + u_n \cdot u_{obs} - (u_n \cdot u_{sun})(u_n \cdot \cdot u_{obs})} F_{ref} \tag{4}$$

$$F_{ref} = B_{spec} + (1 - B_{spec})(1 - u_{sun} \cdot u_h)^5 \tag{5}$$

$$F_{diff} = \frac{28B_{diff}}{23\pi}(1 - B_{spec})\left(1 - \left(\frac{u_n \cdot u_{sun}}{2}\right)^5\right)\left(1 - \left(\frac{u_n \cdot u_{obs}}{2}\right)^5\right) \tag{6}$$

$$F = F_{spec} + F_{diff} \tag{7}$$

In equations (4) to (6): F_{spec} represents the specular reflection; F_{diff} is the diffuse reflection; $F_{reflect}$ is the Fresnel reflectance; B_{spec} and B_{diff} represent the Hejduk Weighting Coefficients; n_u and n_v represent the directional distribution of the specular reflection; u_n represents the facet normal vector normalized, represented in Fig. 3 as **n**; u_h represents the normalized half- vector between the light source and observer, represented in Fig. 3 as **h**; u_u represents the local horizontal normalized vector, represented in Fig. 3 as **u**; u_v represent the local vertical normalized vector, represented in Fig. 3 as **v**; u_{sun} represent the normalized sun vector, represented in Fig. 3 as $\mathbf{k}_1$; and u_{obs} represents the normalized observer vector, represented in Fig. 3 as $\mathbf{k}_2$.

The comparison between Phong and Defined BRDF models for computational efficiency and accuracy of light curve inversion is performed to determine the preferred BRDF model for accurate light curve inversion procedure.

2.3. Low-resolution RSO images

The low-resolution images used in this paper are modeled after the Fast Auroral Imager (FAI), a camera module onboard the CASSIOPE satellite, NORAD ID 39265. Its purpose is to observe the auroral activities on Earth and has very similar properties compared to commercial-grade star trackers onboard typical LEO satellites. The camera has an imaging frequency of one image per second, during which it captures infrared lights within 630 mm to 1100 mm wavelength range. Images generated are 256 × 256 pixels in dimension and has a fundamental resolution of 0.101 degree. For the purpose of light curve analysis, the reflections of RSOs appear as multiple lit-pixels and clearly observable, representative of typical low-resolution images from commercial-grade star-tracker in low-earth orbit. More information on the FAI sensors as well as the available images can be also found at:

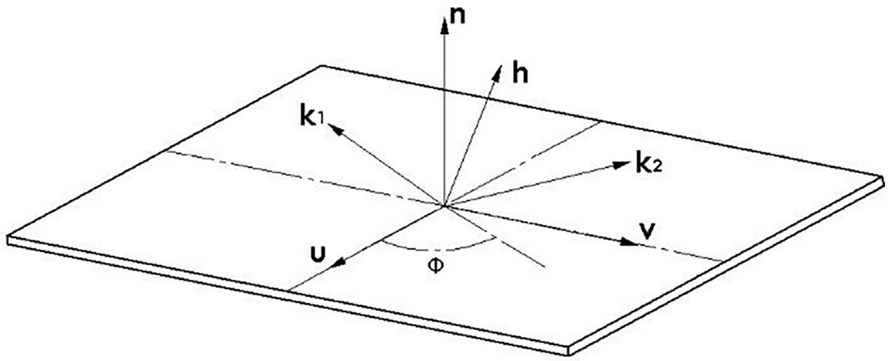

Fig. 3. Observation Geometry for Phong Facet Model.

(University of Calgary, Department of Physics and Astronomy, 2019).

The specific sequences of images used for this paper were taken between July 15*th*, 2019, to July 19*th*, 2019. 6 unique sequences were found in the seen from the FAI during this time, the start point of the sequences and the duration of each of the sequences is shown in Table 4. With the observations coming right after the launch all 3 RCM satellites (NORAD IDs 44,322 – 44324) retain their distinctive formation and move across the sequence in tandem. The latest RCM satellites (Fig. 4) were launched in June 2019, following the highly successful RADARSAT-1, launched by NASA in 1995, and RADARSAT-2, in 2007.

RCM satellites have a unique geometry with a rectangular prism bus, large solar panel, and large SAR panels. To generate an accurate 3D model of RCM only the largest facets were chosen to be modeled as shown in Fig. 5. The smaller features, such as surface irregularities or patch antennas, are approximated by there effect on the optical parameters of the larger facet. Simplifying the complex 3D shape into the largest facets simplifies the problem allowing the light curves to be easily generated with a facet model, with the draw back of possibly missing contributions from small components. In future work the generation of RSOs 3D models should look into the trade off between accuracy of the facet model and the smallest facet modeled. Note that the sequences that were chosen because their path is close to center of the field of view (FOV) of FAI, compared to later sequences which captures off-centered paths, resulting in a shorter and comparatively

Table 2
Particle Swarm Optimization Parameters **Particle Swarm Optimization Parameters.**

Total Iterations	100
Number of Points	20,000
Max Velocity Attitude	30
Max Velocity Phong Parameters	1000
Max Velocity Standard Parameters	0.5
Current Velocity Weight	0.9
Local Optimum Velocity Weight	1.2
Global Optimum Velocity Weight	1.2
Beta	0.99
Cooling Factor	5

Table 3
Fast Auroral Imager Parameters.

Fast Auroral Imager Parameters	
FOV	26 degrees
Aperture Diameter	1.7 cm
Focal Length	6.89 cm
Integration Time	0.1 s
Image Frequency	1 Hz
Spectral Band	650 – 1100 nm
Pixels	256 × 256
Pixel Size	26 μm
Quantum Efficiency	0.66

Table 4
Sequence Star Time and Duration.

Sequence Start Time and Duration	
2019–06–15 23:35:49	27 Seconds
2019–06–16 01:14:33	40 Seconds
2019–06–16 02:53:16	33 Seconds
2019–06–17 12:37:11	37 Seconds
2019–06–17 14:16:03	34 Seconds
2019–06–17 15:54:44	42 Seconds

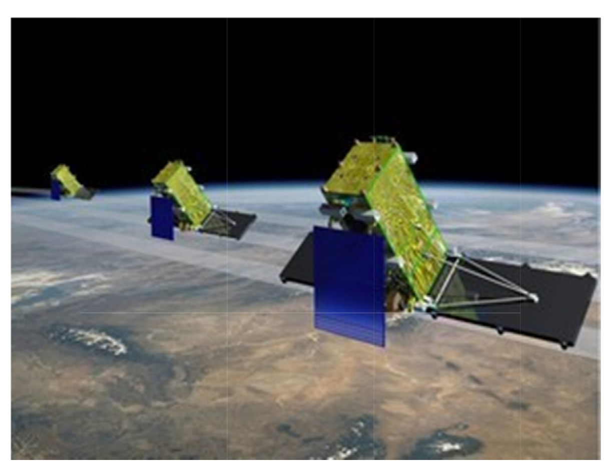

Fig. 4. The RCM, (RADARSAT Constellation Mission) Image Credit: MacDonald, Dettwiler and Associates ltd.(MDA).

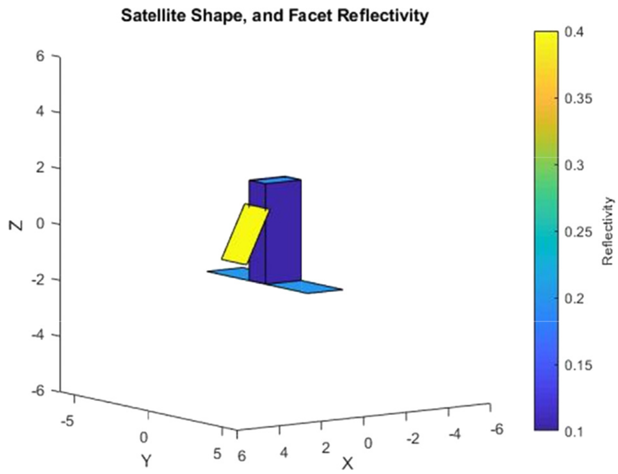

Fig. 5. The RCM, (RADARSAT Constellation Mission) Facet Model.

less ideal sequence due to there being less data points. Sample images of RCM constellation seen from FAI and equivalent simulated images are illustrated in Fig. 1.

2.4. Simulation parameters

Due to the lack of access to real world data for RCM's attitude information the tests performed in the paper were all done in the simulated environment SBOIS outlined in (Clark, et al., 2020), (Clark, et al., 2021), (Clemens, 2019) with a flowchart of optimization method shown in Fig. 6. The optimization algorithm, the same implementation of particle swarm optimization was used as in (Clark, et al., 2020), with the number of iterations and parameters in Table 2. There are three unique tests being performed on the simulated data a) Attitude only, b) Optical Property only, c) Attitude and optical property estimates. In each unique test all the other information is known with only the parameter being analyzed being unknown. Each of the test will run a sequence 10 times in a simulated environment with different observed attitude and optical properties with the results being averaged to get the results for each test. The simulated optical properties are randomly generated and constrained from physical limitations. One example of this is the generated reflectivity number needs to be between 0 and 1 as it is impossible to have a reflectivity less than zero or greater than one. The simulated attitude sequences are generated from a cubic function with a random constrained coefficients. The constraints limit the angular speed to a maximum magnitude of 5 degrees per second and the maximum angular acceleration to 0.5 degree per second. From the results of the three unique tests, a comparison was performed on the convergence rate to an optimal brightness estimates as well as the accuracy of the optimal estimated parameters.(See Table 5)

3. Results

The results from the (a) Attitude, (b) Optical, and (c) Optical and Attitude estimate tests are shown in Table 5,

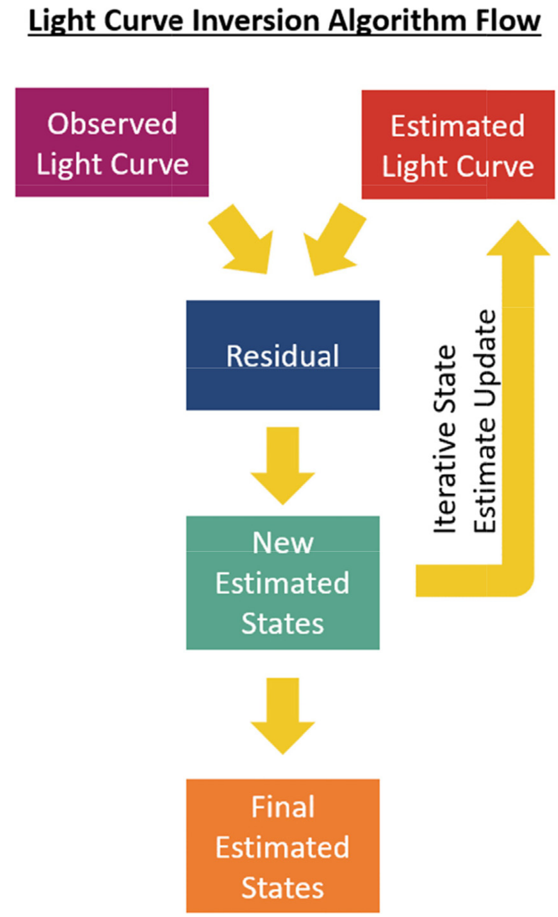

Fig. 6. The flow of the Light Curve Inversion Algorithm.

Table 5
Simulated Attitude Estimation.

Parameter	Phong Model	Standard Model
Average Best Brightness RMSE (Relative Magnitude)	3.6×10^{-7}	2.2×10^{-6}
Average Best Attitude RMSE (Degrees)	1.6	1.2
Average Computation Time (Seconds per Iteration)	11	8.2
Iteration Convergence Rate (#Iterations per Half Life)	10	10
Convergence Rate (Seconds per Half Life)	108	81

Table 6, and Table 7, respectively. Fig. 7 and Fig. 8 illustrate only the convergence rate for optical estimation. Each of the figure of merits used in the comparison of the models are also discussed below.

The first figure of merit used in all three tests is the Average Best Brightness RMSE (ABB RMSE), defined as the root mean squared error between the best estimated light curve and the true light curve. ABB is then averaged across each of the true values and detections. Secondly, computation time is used as a second figure of merit, which includes the average computation time, the iteration convergence rate, and the convergence rate. The average computation

Table 6
Simulated Optical Parameters Estimation.

Parameter	Phong Model	Standard Model
Average Best Brightness RMSE (Relative Magnitude)	2.2×10^{-5}	1.7×10^{-3}
Specular Weighting RMSE (Reflectivity)	7.1×10^{-3}	9.6×10^{-3}
Diffuse Weighting RMSE (Reflectivity)	7.1×10^{-3}	9.6×10^{-3}
Reflectance RMSE (Reflectivity)	N/A	5.4×10^{-3}
Nu RMSE	1.1×10^{3}	N/A
Nv RMSE	1.2×10^{3}	N/A
Optical RMSE (Percent)	2.1	23
Average Computation Time (Seconds per Iteration)	3.7	6.7
Iteration Convergence Rate (#Iterations per Half Life)	8.5	1.5
Convergence Rate (Seconds per Half Life)	32	10

Table 7
Simulated RSO Parameter Estimation Test Results.

Parameter	Phong Model	Standard Model
Average Best Brightness RMSE	0.51	0.27
Average Best Attitude RMSE	5.3	0.66
Specular Weighting RMSE (Reflectivity)	8.8×10^{-4}	5.0×10^{-3}
Diffuse Weighting RMSE (Reflectivity)	8.8×10^{-4}	5.0×10^{-3}
Reflectance RMSE (Reflectivity)	N/A	2.2×10^{-2}
Nu RMSE	4.5×10^{3}	N/A
Nv RMSE	3.3×10^{3}	N/A
Optical RMSE (Percent)	0.9	8.5
Average Computation Time (Seconds per Iteration)	34	27
Iteration Convergence Rate (#Iterations per Half Life)	277	160
Convergence Rate (Minutes per Half Life)	159	73

time is the average time taken to perform on iteration of parameter estimation for the given model and test. The iteration convergence rate refers to the number of iterations it takes for the ABB RMSE to drop by one half-life (drop by 50 %). The convergence Rate is defined as the average time it takes to converge one half-life. The third figure of merit, Average Best Attitude RMSE (ABA RMSE), is used to compare the different estimated attitudes. ABA RMSE is the RMSE of the yaw, pitch, roll residual which is then averaged across all true values and detections. For the optical figure of merit, due to the different scales and parameter used for Phong and Defined model, a direct comparison is not possible. Instead, a mean absolute percentage error (MAPE) is produced for each test to give a comparable figure of merit. MAPE is a standard technique in statistical analysis and regression models (Guo, et al., 2019). This is calculated by taking the absolute present error of the optical parameters then averaging it across all true values and detections.

Attitude optimal estimates in both the attitude, as well as, attitude and optical properties tests are optimized for each sequence independently. This is due to the assumption that the attitude changes enough between sequences that the attitude from a previous sequence has little effect on the attitude of the current sequence. In the current research study this assumption is applicable as this research focuses on the comparison of the two different facet models. Optical properties optimal estimates in both the optimal properties, as well as, attitude and optical properties are optimized for each satellite across all 6 sequences. This is possible as the optical properties of the target RSO (RCM 1, 2 and 3 in this case) are assumed to be constant across the few days that the sequences were taken on.

The results from the attitude only simulated test shows that the Phong model outper- formed the Defined model in ABB RMSE, but not in ABA RMSE. These FOM indi-

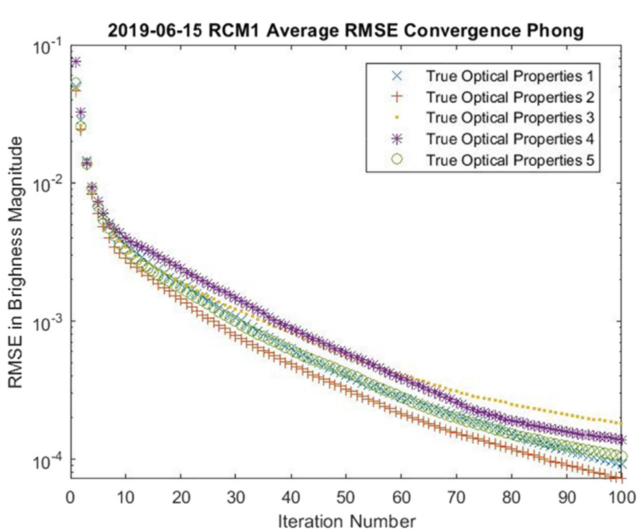

Fig. 7. Convergence Rate of Optical Parameters for the Phong Model.

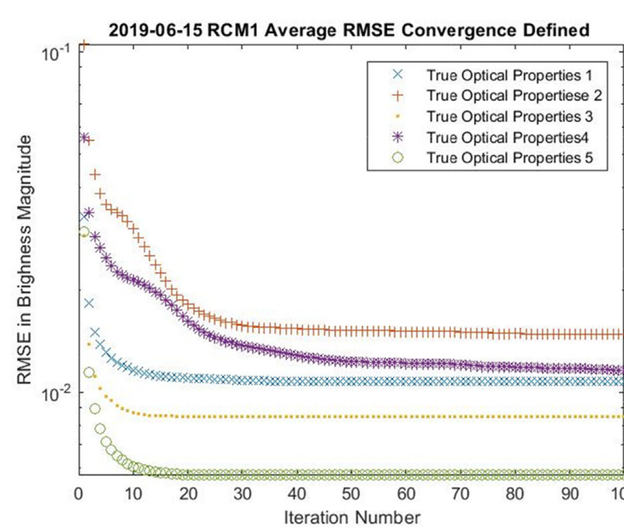

Fig. 8. Convergence Rate of Optical Parameters for the Defined Model.

cates that the Phong model's complex design space leads to the particle swarm optimization algorithm to converge better on local optimum but not on the global optimum which is represented by the true attitude. The Phong and Defined models have approximately the same iteration convergence rate the Phong model has a longer average computation time compared to the Defined model. In summary, the Defined model yields more accurate attitude estimates in less time; thus, it is a preferred modeling technique over the Phong model with particle swarm optimization. Just like the Defined model, the Phong model performance changes based on the optimization algorithm, indicating that implementing other optimization algo- rithms may lead to improved performance both in accuracy and computation speed.

The results from the optical parameter estimates show that the Phong model had better convergence in ABB magnitude, but not as good Optical MAPE as the defined model. This is similar to the attitude only tests, the results showed greater convergence to local minimum, but had problems converging to the global optimum. By comparing the optical parameter results, the same trend can be seen with the Defined and Phong model. The MAPE for all optical parameters is 12 % for the Defined model, compared to the Phong model, which has a 53 % MAPE. The Phong model requiring 5 inputs versus the Defined models 4 leads to a larger response surface that will contain more local optimum. The effect of the large response surface is not just reflected in the accuracy, but also in the iteration convergence rate between the two models, with the Defined model converging about 5.5 times faster. It is observed from the convergence rates (Fig. 7 and Fig. 8) that the Defined model converges before 20 iterations, whereas it takes more than 80 iterations for the Phong model. The results from the attitude and optical parameter estimation test (Table 6) show that increased number of parameters greatly reduces the ABB RMSE convergence for both mod- els. The Defined model performed better than the Phong model in ABA RMSE; however, the two methods have different convergence rates compared to the attitude estimation test. The difference in convergence rates is attributed to the defined model converging quicker and having a slightly faster computation time. Comparing the attitude to attitude and op- tical parameter test shows that the Defined model had a decrease in ABA RMSE, while the Phong model had an increase of ABA RMSE. The increase in ABA RMSE was predictable in the Phong model as the response surface greatly increased in size. In comparison, the ABA RMSE in the Defined model decreased with the increase in response surface. It is theorized with the Defined model converging quicker than the Phong model, the increase in parameters allows for more sensitivity in the solution and a larger number of sub-optimal solutions being rejected in the optimization process. If the Phong model was run through more iteration a similar trend should emerge.

Both methods performed well in determining the optical properties and attitude; the Defined model determining the ABA with less than one degree RMSE across the full sequence of images. The comparison of the optical parameter MAPE has 11 % accuracy for the Defined model and 28 % for the Phong model. The Defined model outperformed the Phong model, which matches the results from the optical property and attitude only test. The largest change in the parameter test is observed when looking at the convergence rate, which increased over an order of magnitude compared to the attitude and optical parame- ters only tests. The increase in parameterization accounts for a majority of the increased iteration convergence rate, which is largely attributed to the local optimum distribution becoming more complex with the more parameters. Another factor in larger parameteriza- tion is that more information needs to be calculated and updated each time, leading to (at best) approximately 2 times increase in the computation time for both models. The Defined model outperformed the Phong model in both the convergence rate and computation time, allowing it to converge around twice as fast as the Phong model. With better accuracy and computation speed, the Defined model was found to perform better than the Phong model for parameter estimation of RSO's in the current implementation for light curves from low resolution space based imagers.

Comparing the proposed RSO attitude and optical property estimation technique to other techniques the strengths of the currently implemented models and areas to further research can be seen. The (Guo, et al., 2019) implementation of light curve inversion follows a similar methodology, as well as the results (Wetterer & Jah, 2009). In this publication, the attitude is determined from the orthographic projection of the target with an iterative attitude estimation. This algorithm was implemented in a lab set up with a known target and a distinct and non-homogeneous circular pattern; the results for single image attitude estimation shows that the attitude of the object can be found to within 1 degree. These results are similar to the ones found in the paper with two large differences between the methodologies. First, the iterative method is used for singular images, not requiring image sequences. The driving factor behind the ability to only use one image for attitude estimation is the second major difference between these two methodologies. The second difference is that the target object is a significant portion of the FOV, unlike with low resolution RSO detections which are commonly subpixel or few pixel-sized objects. The smaller size of RSO's in low resolution space based images limits the applicability that this method has in this papers context. Having both a relatively high spatial resolution and low spatial resolutions sensor for attitude determination of RSO's can lead to leveraging the strength of both types of sensors in current SSA applications, leading to better estimated attitude states.

Comparing more similar methodologies for light curve inversion techniques we look at results such as presented in (Wetterer & Jah, 2009) (Pesce, et al., 2019). A similar methodology was followed to methodology presented in this paper, but with the implementation of an unscented Kalman filter (UKF) as an optimization algorithm. In (Wetterer & Jah, 2009) a simplified Cook and Torrance reflectance model is used in contrast with the Defined model used in this paper. The target objects were rocket bodies, which give easy shapes for simulated environment tests. The simulated environment tests give preliminary results that the UKF and light curve model was able to estimate both the attitude and optical properties within reasonable accuracy. When moving to real data, it was found that the simplified reflection model would not be sufficient for the more complex RSO shapes. The Defined model implemented in this paper has shown the ability in a simulated environment to overcome the higher fidelity model requirements. The implementation of the UKF shows impressive results with more complex methods, such as Multiplicative Extended Kalman Filter (MEKF) being implemented in (Pesce, et al., 2019) and (Ashikhmin & Shirley, 2000). The power of the UKF and other Kalman filtering methods, highlighted in (Wetterer & Jah, 2009) (Pesce, et al., 2019), should be compared to the currently implemented optimization algorithm (PSO) to see the accuracy and computation speed differences between the two.

In a recent study performed by (Pesce, et al., 2019), a comparison of different filter techniques for state estimation of uncooperative objects was performed on MEKF and different minimum energy filter models. The results showed similar performance on the estimation of targets attitude from both MEKF and a second order minimum energy filter. The MEKF was able to estimate attitude with a transient error RMSE of 1.21 degrees and steady state error RMSE of 0.35 degrees, with a given priory and inertial matrix of the target. With no priory estimate and inertial matrix, it was mentioned that the second order minimum energy filter does perform better, giving an error rate of less than one degree after convergence. Comparing the results found from this paper shows that the method for attitude estimation from light curve inversion and PSO give similar accuracy as other currently used techniques. From (Pesce, et al., 2019), it can also be seen that the choice of state estimator greatly effects the accuracy of the attitude estimate depending on the scenario and information available to the filter. The results of different state estimators means that it is highly unlikely there will be one optimum RSO attitude estimation algorithm for all situations. Instead, it is much more probable that the optimization algorithms will be chosen based on the spatial resolution, sensitivity, and temporal resolution available, much like with orbital propagators. Identifying which algorithms perform optimally in different scenarios will allow for better attitude estimation and high-fidelity attitude estimation. Some of those scenarios include terrestrial verses space-based detections, low vs high spatial resolution cameras, and tracking vs stare mode.

4. Conclusion and future work

We have compared the Defined model with the Phong model of an RSO for three types of parameter estimation using low-resolution images in a simulated environment. The re- sults indicate that, with particle swarm optimization, the Defined model outperforms the Phong model in almost all metrics; accuracy, computation time, and convergence rate. The Phong model was examined due to the directionality of incoming light effecting the specular and diffuse reflections off the facet, which is not considered in the Defined model. It was expected to increase the accuracy of the estimated states due to the additional parameters (n_u, n_v), reducing the number of geometries that provide the same brightness. Instead, the results indicate that the increase in number of parameters exponentially increased the size of the response surface, making convergence to the local optimum require more iterations. Furthermore, the increase in response surface leads to more local optima present when com- pared to the Defined model. The increase in number of local optima leads to the Phong model requiring more iterations and data points, making it significantly more computationally intensive to achieve a similar accuracy. The implementation of different, more intelligent optimization algorithms in the future could lead to an increase in performance of the Phong model.

In conclusion, the Defined model form the foundation in characterization of unidentified RSO attitude and optical properties. In the future this technology will lead to better remote sensing of RSO's health and operation status. Future work looks to take this method from being performed in a simulated environment to using real data. To perform a real world feasibility test the light curve analysis for attitude estimation and optical characterization will be performed using on-orbit images of operational spacecraft with known health status. Currently this is being performed on NEOSSAT images from their Early Orbit Phase (EOP) taken from a ground-based telescope provided by DRDC Canada. While ideally the images would be from an space based sensor, due to the lack of labeled space based images with auxiliary information ground based observations were chosen. By comparing the results from the proposed estimation technique to on-orbit NEOSSAT data (truth data), we can validate the accuracy of the proposed light curve inversion methodology. The current methodology still has area of improvement that will be further enhanced such as: including intelligent optimization, improved BRDF modeling, and efficient parameterization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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